REVISION TEST 1

1. Please check that this question paper contains 12 printed pages.
2. Please check that this question paper has 45 questions.
3. This paper is divided into three sections- Mathematics, Physics and Chemistry.
4. Please use a dark and legible pencil to darken and write the answer.
5. Each section carries 60 marks.
6. Each section contains 15 questions divided into two sub-sections.
7. Section 1 contains 10 Multiple Choice Questions and Section 2 contains 5 Integer type questions.

MATHEMATICS

Total positive marks: _____
Total negative marks: _____
Total Marks: _____

Section 1
Maximum Marks: 40

1. This section contains 10 questions.
2. For each question, choose the correct option by darkening the circle next to it.
3. Four marks are awarded for every correct answer. No marks are awarded if question is left unattempted and -1 marks are added if answer chosen in incorrect.

1. A particle moves along the parabola $y = x^2$ in the first quadrant in such a way that its x-coordinate (measured in metres) increases at a rate of 10 m/sec. If the angle of inclination θ of the line joining the particle to the origin changes, when $x = 3$ m, at the rate of k rad/sec, then the value of k is

   ◯ 1 ◯ 2 ◯ 1/2 ◯ 1/3

2. Let $L = \lim\limits_{x\to 0} \dfrac{xe^{\alpha x} - \beta \sin x}{x^3}$ is finite. Consider the following two statements.
   S1: $\alpha^{99} + \beta^{99} = -1$
   S2: The value of L is $\dfrac{1}{6}$.
   - ◯ S1 is true and S2 is false.
   - ◯ Both S1 and S2 are false.
   - ◯ Both S1 and S2 are true.
   - ◯ S1 is false and S2 is true.

3. Let $f(x) = \begin{cases} \dfrac{ax^2 + 2ax + 3}{4x^2 + 4x - 3} & x \neq -\dfrac{3}{2}, \dfrac{1}{2} \\ b & x = -\dfrac{3}{2}, \dfrac{1}{2} \end{cases}$ be continuous at $x = \dfrac{-3}{2}$. If $fof(x) = \frac{7}{5}$, then x is equal to
   ◯ 1 ◯ 0 ◯ 1.4 ◯ 2

4. Let $f(x) = x^3 + bx^2 + cx + d;\ 0 < b^2 < c$, then $f(x)$
   - ◯ Is strictly increasing
   - ◯ Has local maxima
   - ◯ Has local minima
   - ◯ Is bounded curve

5. If $\lim\limits_{x\to\infty} x\left(\tan^{-1}\left(\dfrac{x+\lambda}{x+\ \mu}\right) - \dfrac{\pi}{4}\right) = 1$, then ordered pair(s) (λ, μ) can be
   ◯ (2022, 2023) ◯ (2023, 2022) ◯ (2024, 2022) ◯ (2022, 2024)

6. Let $f(x) = x^3 + 6x^2 + px + 2$, if the largest possible interval in which $f(x)$ is decreasing function is $(-3, -1)$, then p equals :
   ◯ 3 ◯ 9 ◯ -2 ◯ None of these

7. A spherical balloon is filled with 4500π cubic metres of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic metres per minute, then the rate (in metres per minute) at which the radius of the balloon decreases 49 minutes before the leakage began is :
   ◯ $\dfrac{6}{7}$ ◯ $\dfrac{4}{9}$ ◯ $\dfrac{2}{9}$ ◯ $\dfrac{9}{2}$

8. For the function $f(x) = e^{\sin|x|} - |x|,\ x \in \mathbb{R}$, consider the following statements :
Statement I: f is differentiable for all $x \in \mathbb{R}$.
Statement II: f is increasing in $\left(-\pi, \frac{\pi}{2}\right)$.
In the light of the above statements, choose the correct answer from the options given below :

   ◯ Both Statement I and Statement II are true.
   ◯ Both Statement I and Statement II are false.
   ◯ Statement I is true but Statement II is false.
   ◯ Statement I is false but Statement II true.

9. Let $f(x) = \begin{cases} \frac{1}{3} & x \leq \frac{\pi}{2} \\ \frac{b(1-\sin x)}{(\pi-2x)^2} & x > -\frac{\pi}{2} \end{cases}$. If f is continuous at $x = \frac{\pi}{2}$, then the value of $\int_0^{3b-6} |x^2 + 2x - 3| dx$ is :

   ◯ 5 ◯ 2 ◯ 3 ◯ 4

10. The lateral edge of a regular hexagon pyramid is 1 cm. If the volume is maximum, then its height must be equal to

   ◯ $\frac{1}{3}$ ◯ $\frac{2}{3}$ ◯ $\frac{1}{\sqrt{3}}$ ◯ 1

Section 2
Maximum Marks: 20

1. This section contains 5 questions.
2. For each question, write the integer answer in the blank provided.
3. Four marks are awarded for every correct answer. No marks are awarded if the question is left unattempted, and -1 mark is added if the answer chosen is incorrect.

11. Let C be a curve defined by $y = e^{a+bx^2}$. If the curve C passes through the point $P(1,1)$ and the slope of the tangent at P is (-2), then the value of $|ab|$ is _____.

12. Let [.] denote the greatest integer function and $f(x) = \lim\limits_{n\to\infty} \frac{1}{n^3} \sum\limits_{k=1}^{n} \left[\frac{k^2}{3^x}\right]$. Then, $12 \sum\limits_{j=1}^{\infty} f(j)$ is equal to _____.

13. Number of points where $f(x) = x^2 - |x^2 - 1| + 2||x| - 1| + 2|x| - 7$ is non-differentiable is _____.

14. If the distance of the point on $y = x^4 + 3x^2 + 2x$ which is nearest to the line $y = 2x - 1$ is p. Find $5p^2$. _____

15. If absolute maximum value of $f(x) = \dfrac{1}{|x|+5} + \dfrac{1}{|x-5|+5}$ is A, then $10A$ is _____

PHYSICS

Total positive marks: _____
Total negative marks: _____
Total Marks: _____

Section 1
Maximum Marks: 40

1. This section contains 10 questions.
2. For each question, write the integer answer in the blank provided.
3. Four marks are awarded for every correct answer. No marks are awarded if the question is left unattempted, and -1 mark is added if the answer chosen is incorrect.

16. A current I flows through a thin wire shaped as a regular polygon of n sides which can be inscribed in a circle of radius R. The magnetic field induction at the centre of the polygon due to one side of the polygon is

◯ $\dfrac{\mu_0 I}{\pi R}\left(\tan\dfrac{\pi}{n}\right)$ ◯ $\dfrac{\mu_0 I}{4\pi R}\left(\tan\dfrac{\pi}{n}\right)$ ◯ $\dfrac{\mu_0 I}{2\pi R}\left(\tan\dfrac{\pi}{n}\right)$ ◯ $\dfrac{\mu_0 I}{2\pi R}\left(\cos\dfrac{\pi}{n}\right)$

17. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material:
Assertion (A): The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains.
Reason (R): At high enough temperature, the domain structure of ferromagnetic material disintegrates. Thus, magnetization will disappear at high enough temperature known as Curie temperature.
In the light of the above statements, choose the correct answer from the options given below:

◯ (A) is true but (R) is false.

◯ Both (A) and (R) are true but (R) is not the correct explanation of (A).

◯ Both (A) and (R) are true and (R) is the correct explanation of (A).

◯ (A) is false but (R) is true.

18. A straight long wire carries a current $I_1 = 10$ A. It repels another parallel wire with a force of 10^{-5} N/m, when the distance between the two wires is 10 cm. A third parallel wire carrying current $I_3 = 2$ A is placed parallel to the two wires, midway between them. All three wires in a plane. The net magnetic force on 1 m length of the wire is,

◯ 4.8×10^{-5} N/m ◯ 9.6×10^{-5} N/m ◯ 2.4×10^{-5} N/m ◯ 3.6×10^{-5} N/m

19. An electron and a proton with equal momentum enter perpendicularly into a uniform magnetic field, then

◯ the path of proton shall be more curved than that of electron.

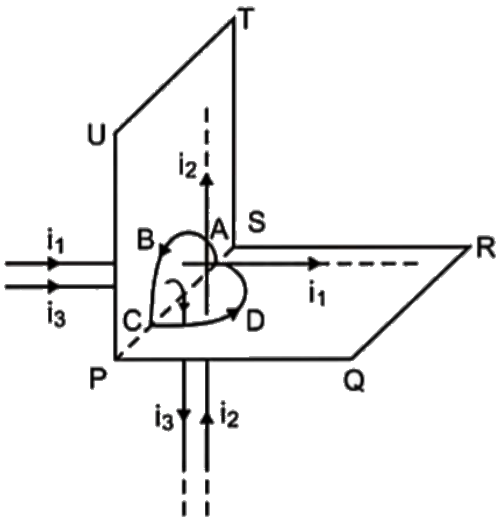
◯ the path of proton shall be less curved than that of electron.

◯ both are equally curved.

◯ paths of both will be straight line.

20. The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibility $\chi_{Mg} = 1.2 \times 10^{-5}$) is : ◯ $\frac{5}{3} \times 10^{-5}$ % ◯ $\frac{5}{6} \times 10^{-4}$ % ◯ $\frac{5}{6} \times 10^{-5}$ % ◯ $\frac{6}{5} \times 10^{-3}$ %

21. Figure shows an Amperian path $ABCDA$. Part ABC is in vertical plane $PTSU$ while part CDA is in horizontal plane $PQRS$. Direction of circumlation along the path is shown by an arrow near point B and at D. $\oint \vec{B} \cdot d\vec{l}$ for this path according to Ampere's Law will be :



◯ $(i_1 - i_2 + i_3)\mu_0$ ◯ $(-i_1 + i_2)\mu_0$ ◯ $i_3\mu_0$ ◯ $(i_1 + i_2)\mu_0$

22. In region $x > 0$, a uniform and constant magnetic field $\vec{B}_1 = 2B_o\hat{k}$ exists. Another uniform and constant magnetic field $\vec{B}_2 = B_0\hat{k}$ exists in region $x¡0$. A positively charged particle of mass m and charge q is crossing origin at time $t = 0$ with a velocity $\vec{u} = u_o\hat{i}$. The particle comes back to its initial position after a time : (B_0, u_o are positive constants)

◯ $\frac{3\pi m}{2qB_0}$ ◯ $\frac{2\pi m}{qB_0}$ ◯ $\frac{3\pi m}{qB_0}$ ◯ particle does not come back to its initial position

23. A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2 m long and 0.2 m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is :

◯ 1 A/m ◯ 2.4×10^{-3}A/m ◯ 1.2×10^{3}A/m ◯ 2.4×10^{3}A/m

24. A galvanometer has a resistance of 50 Ω and gives a full-scale deflection for a current of 2 mA. To convert it into an ammeter capable of measuring currents up to 2 A, the shunt resistance required is closest to:

◯ 0.05 Ω ◯ 0.50 Ω ◯ 0.02 Ω ◯ 0.11 Ω

25. The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}$ cm is 2 A. The magnetic field at its centroid is $\alpha \times 10^{-5}$ T. The value of α is _____.

◯ $2\sqrt{3}$ ◯ $\frac{\sqrt{3}}{2}$ ◯ $3\sqrt{3}$ ◯ $\sqrt{3}$

Section 2
Maximum Marks: 20

1. This section contains 5 questions.
2. For each question, write the integer answer in the blank provided.
3. Four marks are awarded for every correct answer. No marks are awarded if the question is left unattempted, and -1 mark is added if the answer chosen is incorrect.

26. A short bar magnet placed with its axis at 30° with an external field of 800 Gauss, experiences a torque of 0.016 N.m. The work done in moving it from most stable to most unstable position is $\alpha \times 10^{-3}$ J. The value of α is _____

27. A iron rod of volume 10^{-4} m^3 and relative permeability 1000 is placed inside a long solenoid wound with 5 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment (in Am^2) of the rod is _____

28. A 5 mg particle carrying a charge of $5\pi \times 10^{-6}$ C is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}$ m/s in a region having magnetic field $\vec{B} = 0.1\hat{k}$ Wb/m^2. It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is _____.

29. The magnetic field at the centre of a current-carrying loop of radius 0.1 m is $5\sqrt{5}$ times that at a point along its axis. What is the distance (in m) of this point from the centre of the loop?

_____

30. An electron is shot into one end of a solenoid. As it enters the uniform magnetic field within the solenoid, its speed is 800 m/s and its velocity vector makes an angle 30° with the the central axis of the solenoid. The solenoid carries 4.0 A current and has 8000 turn along its length. The number of revolutions made by the electron within the solenoid by the time it emerges from the solenoid's opposite end is _____$\times 10^3$. ($\frac{e}{m} = \sqrt{3} \times 10^{11}$C/kg)

CHEMISTRY

Total positive marks: _____
Total negative marks: _____
Total Marks: _____

Section 1
Maximum Marks: 40

1. This section contains 10 questions.
2. For each question, write the integer answer in the blank provided.
3. Four marks are awarded for every correct answer. No marks are awarded if the question is left unattempted, and -1 mark is added if the answer chosen is incorrect.

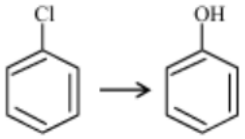
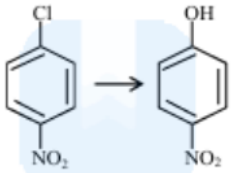
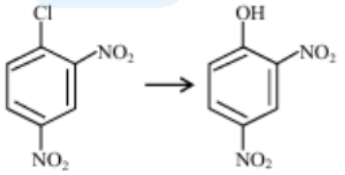
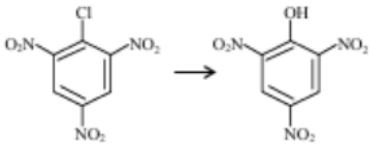
31. $'x'$ is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at 333 K. $'y'$ is the product which is obtained by the reaction of ethane nitrile with methyl magnesium bromide in dry ether followed by hydrolysis. IUPAC name of product obtained from $'x'$ and $'y'$ in the presence of barium hydroxide followed by heating is:

- ◯ 2-Methylpent-4-en-3-one
- ◯ 4-Methylpen-3-en-2-one
- ◯ 4-Methylpent-1-ene
- ◯ 2-Methylpen-3-one

32. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material:
Assertion (A): Alcohols react both as nucleophiles and electrophiles.
Reason (R): Alcohols react with active metals such as sodium, potassium and aluminum to yield corresponding alkoxides and liberate hydrogen.
In the light of the above statements, choose the correct answer from the options given below:

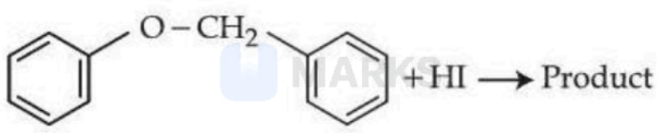
- ◯ (A) is true but (R) is false.
- ◯ Both (A) and (R) are true but (R) is not the correct explanation of (A).
- ◯ Both (A) and (R) are true and (R) is the correct explanation of (A).
- ◯ (A) is false but (R) is true.

33. Match List-I with List-II.

List-I Conversion		List-II Reagents, Conditions used	
(A)		(I)	Warm, H_2O
(B)		(II)	(a) NaOH, 368 K ; (b) H_3O^+
(C)		(III)	(a) NaOH, 443 K; (b) H_3O^+
(D)		(IV)	(a) NaOH, 623 K, 300 atm ; (b) H_3O^+

- ◯ A - II, B - III, C - I, D - IV
- ◯ A - III, B - IV, C - II, D - I
- ◯ A - IV, B - III, C - II, D - I
- ◯ A - IV, B - III, C - I, D - II

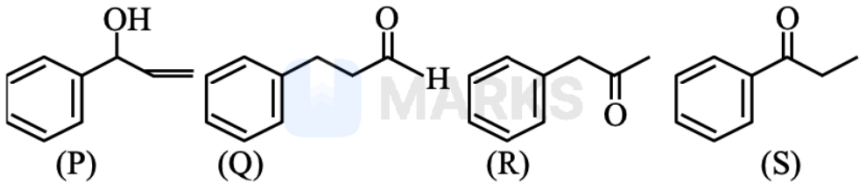
34. Consider the following reaction :



Statement I: In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene.
Statement II: In the above reaction, the $-O-CH_2-$ bond is cleaved to give the product.
In the light of the above statements, choose the correct answer from the options given below :

- ◯ Both Statement I and Statement II are true.
- ◯ Both Statement I and Statement II are false.
- ◯ Statement I is true but Statement II is false.
- ◯ Statement I is false but Statement II true.

35. Given below are four isomeric compounds (P, Q, R, S)



Identify correct statements from below.
A. Q, R and S will give precipitate with 2,4−DNP.
B. P and Q will give positive Bayer's test.
C. Q and R will give sooty flame.
D. R and S will give yellow precipitate with I_2/NaOH.
E. Q alone will deposit silver with Tollen's reagent.
Choose the correct option.

- ◯ A, B, D and E only.
- ◯ A and E only.
- ◯ C and E only.
- ◯ A, C and E only.

36. Given below are two statements :
Statement I: Cross aldol condensation between two different aldehydes will always produce four different products.
Statement II: When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.
In the light of the above statements, choose the correct answer from the options given below:

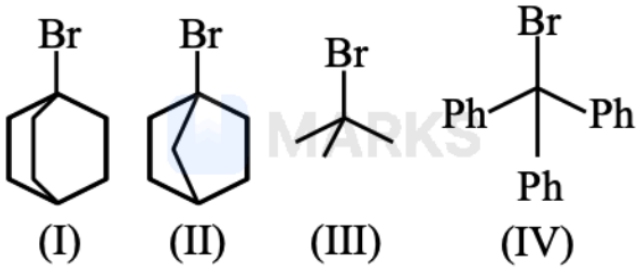
◯ Both Statement I and Statement II are true.

◯ Both Statement I and Statement II are false.

◯ Statement I is true but Statement II is false.

◯ Statement I is false but Statement II true.

37. The correct order of the rate of reaction of the following reactants with nucleophile by SN1 mechanism is: (Given : Structure I and II are rigid)

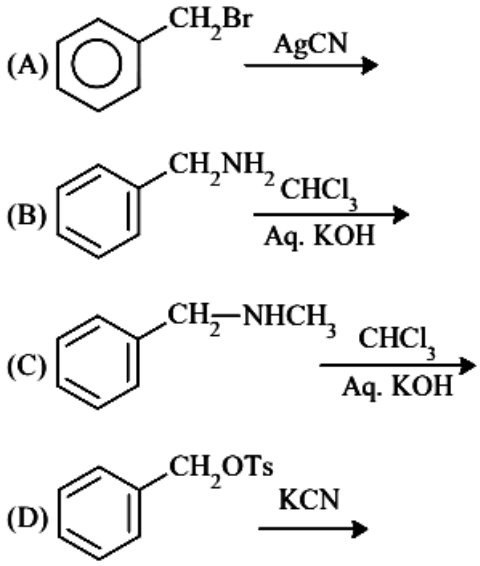


◯ III < I < II < IV ◯ IV < III < II < I ◯ I < II < III < IV ◯ II < I < III < IV

38. Number of molecules from below which cannot give ioddoform reaction is :
Ethanol, Isopropyl alcohol, Bromoacetone, 2-Butanol, 2-Butanone, Butanal, 2-Pentanone, 3-Pentanone, Pentanal and 3-Pentanol

◯ 5 ◯ 4 ◯ 3 ◯ 2

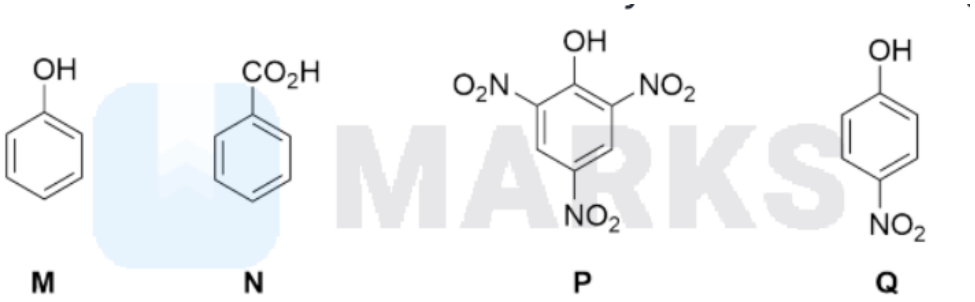
39. Benzyl isocyanide can be obtained from



Choose the correct answer from the options given below:

- ◯ A and B only.
- ◯ A and C only.
- ◯ B and D only.
- ◯ D and E only.

40. What is the correct order of acidity for the following compounds?



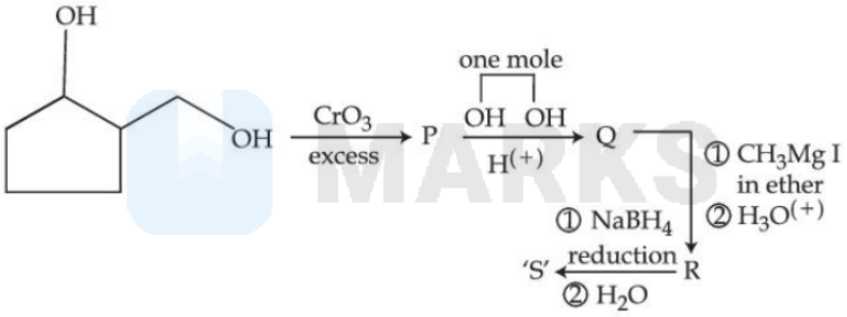
- ◯ $P > N > Q > M$
- ◯ $P > Q > N > M$
- ◯ $N > P > Q > M$
- ◯ $N > P > M > Q$

Section 2
Maximum Marks: 20

1. This section contains 5 questions.
2. For each question, write the integer answer in the blank provided.
3. Four marks are awarded for every correct answer. No marks are awarded if the question is left unattempted, and -1 mark is added if the answer chosen is incorrect.

41. Grignard reagent RMgBr (P) reacts with water and forms a gas (Q). One gram of Q occupies 1.4 dm^3 at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh _____g. (Nearest integer)

42. Consider the reaction sequence:



0.1 mole of a compound 'S' will weigh _____g.

43. 4.7 g of phenol is heated with Zn to give product X. If this reaction goes to 60% completion then the number of moles of compound X formed will be _____$\times 10^{-2}$. (Nearest integer)

44. If all carbonyl isomers of molecular mass (= 86 u) are separately treated with CH_3MgBr followed by acid hydrolysis, how many of them will give racemic mixture? _____.

45. RMgI when reacted ice cold water liberated a gas which occupied 1.4 dm^3/g at STP. The gas produced is further reacted with iodine in the presence of HIO_3 to give compound (X). Compound (X) in presence of Na and dry ether produced compound (Y). Molar mass of compound (Y) is _____g mol^{-1}. (Nearest integer)